

Group Assignment I

Advanced Econometrics 2026–2027

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Question 1

First write out $\sigma_{t+1}^2(\sigma_t^2, \epsilon_t)$, i.e. write the volatility equation as a function of the innovations.

We know:

$$x_t = \mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t$$

and by the given equation for σ_t we know:

$$\sigma_{t+1}^2 = \omega + \left(\alpha + \delta \tanh(-\gamma x_t) \right) \left(\frac{x_t - \mu - \lambda\sigma_t^2}{\sigma_t} \right)^2 + \beta\sigma_t^2.$$

Substituting now gives us:

$$\begin{aligned} \sigma_{t+1}^2 &= \omega + \left(\alpha + \delta \tanh(-\gamma(\mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t)) \right) \left(\frac{(\mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t) - \mu - \lambda\sigma_t^2}{\sigma_t} \right)^2 + \beta\sigma_t^2 \\ &= \omega + \left(\alpha + \delta \tanh(-\gamma(\mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t)) \right) \left(\frac{\sigma_t\epsilon_t}{\sigma_t} \right)^2 + \beta\sigma_t^2 \\ &= \omega + \left(\alpha + \delta \tanh(-\gamma(\mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t)) \right) \epsilon_t^2 + \beta\sigma_t^2. \end{aligned}$$

We have now written $\sigma_{t+1}^2(\sigma_t, \epsilon_t) = \omega + \left[\alpha + \delta \tanh(-\gamma(\mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t)) \right] \epsilon_t^2 + \beta\sigma_t^2$. We want to show that $\{\sigma_t\}_{t \in \mathbb{Z}}$ is SE. We can do this by Bougerol's Theorem.

A1 $\{\epsilon_t\}_{t \in \mathbb{Z}}$ is SE. This holds since $\{\epsilon_t\} \sim TID(v)$ is iid and any iid sequence is SE.

A2 There should exist a finite initialization σ_1^2 such that $\mathbb{E}[\log^+ |\sigma_2^2(\sigma_1^2, \epsilon_1)|] < \infty$. We have:

$$\begin{aligned} \mathbb{E}[\log^+ |\sigma_2^2(\sigma_1^2, \epsilon_1)|] &= \mathbb{E}\left[\log^+ \left| \omega + [\alpha + \delta \tanh(-\gamma(\mu + \lambda\sigma_1^2 + \sigma_1\epsilon_1))] \epsilon_1^2 + \beta\sigma_1^2 \right| \right] \\ &\leq \mathbb{E}\left[\left| \omega + [\alpha + \delta \tanh(-\gamma(\mu + \lambda\sigma_1^2 + \sigma_1\epsilon_1))] \epsilon_1^2 + \beta\sigma_1^2 \right| \right] \quad (\log^+ |z| \leq |z|) \\ &\leq |\omega| + \mathbb{E}\left[\left| \alpha + \delta \tanh(-\gamma(\mu + \lambda\sigma_1^2 + \sigma_1\epsilon_1)) \right| \epsilon_1^2 \right] + |\beta| |\sigma_1^2| \quad (\text{triangle inequality}) \\ &\leq |\omega| + \mathbb{E}\left[(|\alpha| + |\delta| |\tanh(\dots)|) \epsilon_1^2 \right] + |\beta| |\sigma_1^2| \quad (\text{triangle inequality}) \\ &\leq |\omega| + \mathbb{E}\left[(|\alpha| + |\delta|) \epsilon_1^2 \right] + |\beta| |\sigma_1^2| \quad (-1 \leq \tanh(\cdot) \leq 1) \\ &= |\omega| + (|\alpha| + |\delta|) \mathbb{E}[\epsilon_1^2] + |\beta| |\sigma_1^2| \\ &< \infty. \end{aligned}$$

This is finite since $\omega, \alpha, \beta, \delta$ are finite parameters, $\mathbb{E}[\epsilon_1^2] < \infty$ for $\epsilon_1 \sim TID(v)$ with $v > 2$, and σ_1^2 is a finite initialization. So A2 holds.

A3 The contraction holds when

$$\mathbb{E} \left[\log \sup_{x \in \mathcal{X}} \left| \frac{\partial \phi(x, \epsilon_t)}{\partial x} \right| \right] < 0.$$

Here $\phi(\sigma_t^2, \epsilon_t) = \omega + [\alpha + \delta \tanh(-\gamma(\mu + \lambda\sigma_t^2 + \sigma_t\epsilon_t))] \epsilon_t^2 + \beta\sigma_t^2$.

We recall also that $\frac{d \tanh(x)}{dx} = 1 - \tanh^2(x)$.

By the chain rule,

$$\frac{\partial \phi(\sigma_t^2, \varepsilon_t)}{\partial \sigma_t^2} = \delta \varepsilon_t^2 \left(1 - \tanh^2(-\gamma(\mu + \lambda \sigma_t^2 + \sigma_t \varepsilon_t)) \right) \cdot (-\gamma) \left(\lambda + \frac{\varepsilon_t}{2\sqrt{\sigma_t^2}} \right) + \beta.$$

We then bound,

$$\begin{aligned} \left| \frac{\partial \phi(\sigma_t^2, \varepsilon_t)}{\partial \sigma_t^2} \right| &\leq |\gamma| |\delta| \varepsilon_t^2 \left| 1 - \tanh^2(-\gamma(\mu + \lambda \sigma_t^2 + \sigma_t \varepsilon_t)) \right| \left(|\lambda| + \frac{|\varepsilon_t|}{2\sqrt{\sigma_t^2}} \right) + |\beta| \quad (\text{triangle inequality}) \\ &\leq |\gamma| |\delta| \varepsilon_t^2 \left(|\lambda| + \frac{|\varepsilon_t|}{2\sqrt{\sigma_t^2}} \right) + |\beta| \quad (0 \leq 1 - \tanh^2(\cdot) \leq 1) \\ &\leq |\gamma| |\delta| \varepsilon_t^2 \left(|\lambda| + \frac{|\varepsilon_t|}{2\sqrt{\omega}} \right) + |\beta| \quad (\sigma_t^2 \geq \omega \Rightarrow \frac{1}{2\sqrt{\sigma_t^2}} \leq \frac{1}{2\sqrt{\omega}}) \\ &= |\gamma| \left(|\lambda \delta| \varepsilon_t^2 + \frac{|\delta| |\varepsilon_t|^3}{2\sqrt{\omega}} \right) + |\beta|, \end{aligned}$$

where $\sigma_t^2 \geq \omega$ holds for all t since $\alpha > |\delta|$ and $|\tanh(\cdot)| \leq 1$, $\alpha + \delta \tanh(-\gamma x_{t-1}) \geq \alpha - |\delta| > 0$, so the middle term of the σ_t^2 -recursion is non-negative and since $\beta \geq 0$ so is $\beta \sigma_{t-1}^2$. Hence $\sigma_t^2 \geq \omega$.

Therefore,

$$\mathbb{E} \left[\log \sup_{x \in \mathcal{X}} \left| \frac{\partial \phi(x, \varepsilon_t)}{\partial x} \right| \right] \leq \mathbb{E} \left[\log \left(|\gamma| \left(|\lambda \delta| \varepsilon_t^2 + \frac{|\delta| |\varepsilon_t|^3}{2\sqrt{\omega}} \right) + |\beta| \right) \right] < 0,$$

which is exactly the condition given in the question.

Thus, by Bougerol's theorem, $\{\sigma_t^2\}_{t \in \mathbb{Z}}$ converges to a SE process. Since $x_t = \mu + \lambda \sigma_t^2 + \sigma_t \varepsilon_t$ is a measurable function of the jointly SE sequence $\{(\sigma_t, \varepsilon_t)\}_{t \in \mathbb{Z}}$, in this case $\{x_t\}_{t \in \mathbb{Z}}$ is also SE by Krengel's theorem.

Question 2

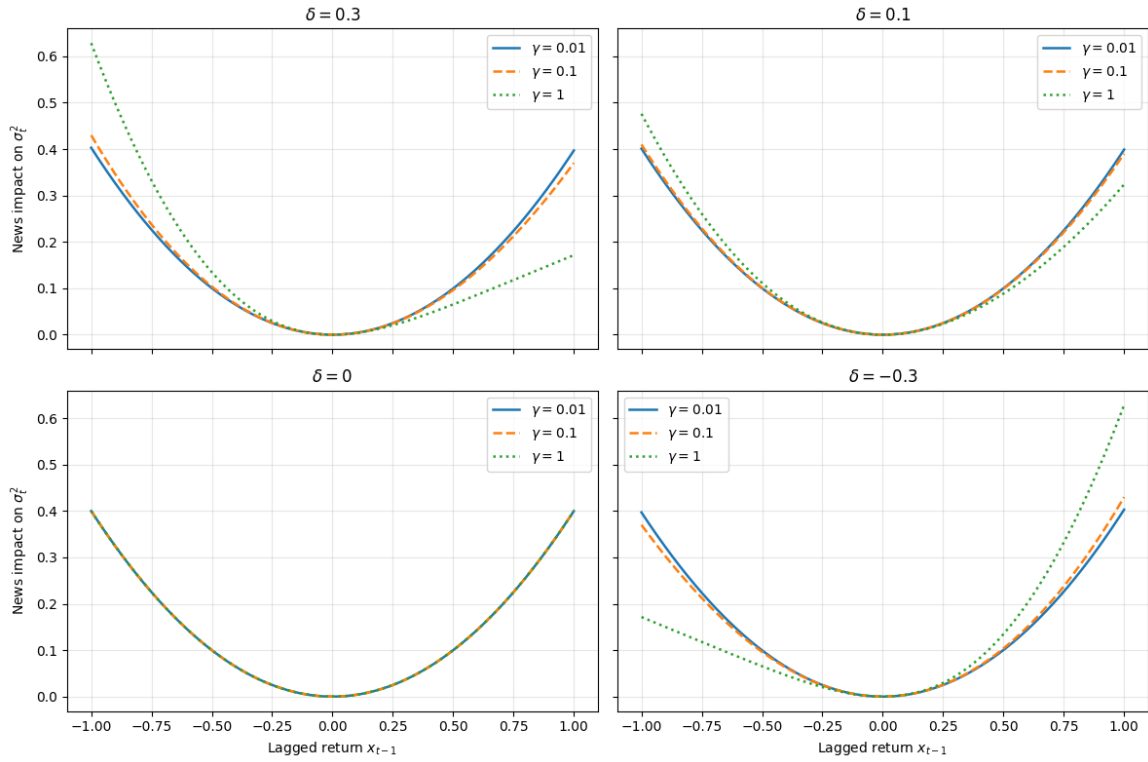


Figure 1: News impact curves of the GARCH-M-L model ($\mu = \lambda = 0$, $\alpha = 0.4$, $\sigma_{t-1}^2 = 1$)

Question 3

Table 1: Descriptive statistics of the daily holding period returns

	PFE	JNJ	MRK	AAPL
Number of observations	3270	3270	3270	3270
Mean	0.041	0.046	0.057	0.107
Median	0.000	0.036	0.046	0.089
Standard deviation	1.368	1.076	1.306	1.783
Minimum	-7.735	-10.038	-9.863	-12.865
Maximum	10.855	7.998	10.408	11.981
Skewness	0.223	-0.150	0.083	-0.076
Excess kurtosis	5.294	9.370	6.637	5.360

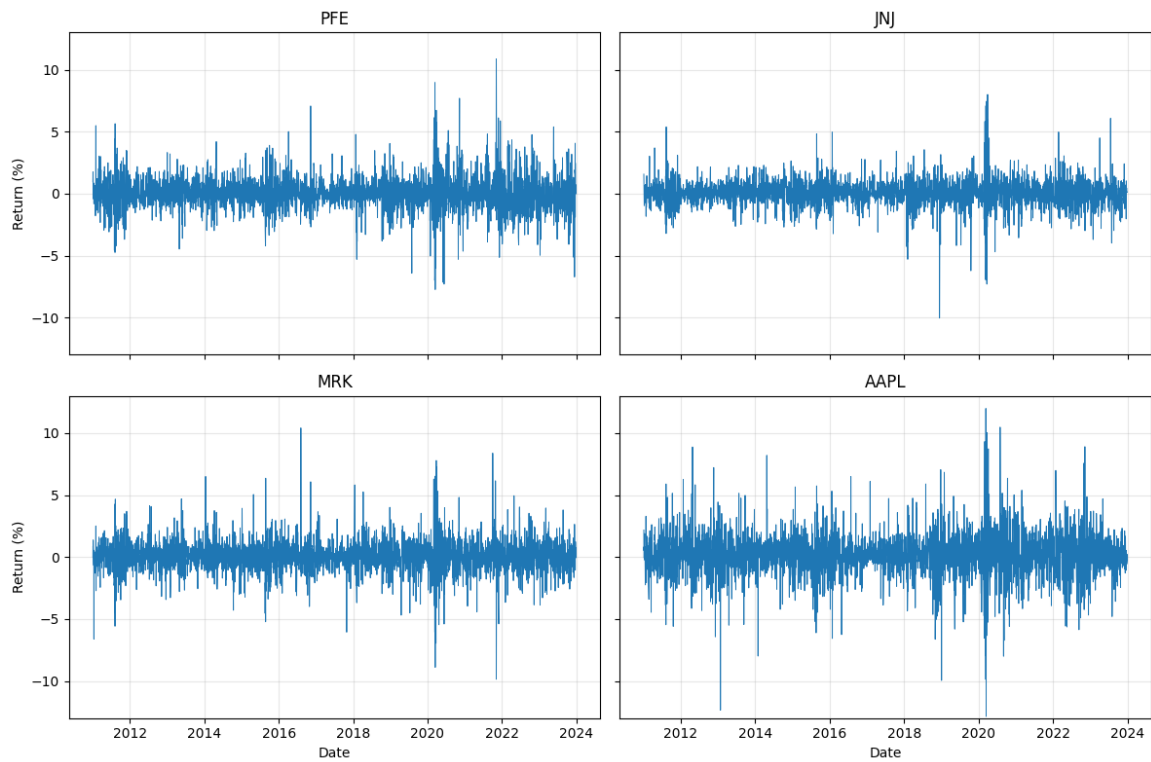


Figure 2: Daily holding period returns of Pfizer, Johnson & Johnson, Merck & Co., and Apple.

Question 4

Table 2: Maximum likelihood estimates of the three volatility models

	PFE			JNJ			MRK		
	M1	M2	M3	M1	M2	M3	M1	M2	M3
μ	0.037	-0.028	-0.031	0.072	0.058	0.033	0.066	0.077	-0.046
λ		0.111	0.104		0.031	0.063		-0.015	0.117
ω	-0.027	-0.028	-0.025	-0.002	-0.002	0.003	0.014	0.015	0.019
α	0.036	0.036	0.043	0.025	0.025	0.026	0.039	0.040	0.040
β	0.961	0.963	0.953	0.923	0.924	0.916	0.906	0.904	0.906
δ			0.029			0.017			0.039
γ			0.410			1.228			4.742
ν	4.651	4.677	4.903	4.394	4.389	4.556	4.498	4.501	4.715
Log-lik	-3769	-3767	-3753	-3283	-3283	-3270	-3893	-3893	-3868
AIC	7549	7546	7522	6576	6577	6557	7796	7798	7752
BIC	7578	7581	7569	6605	6612	6603	7825	7833	7799

Notes: M1, M2, and M3 are denoted as the GARCH, GARCH-M, and GARCH-M-L models, respectively. AIC and BIC are the Akaike information criterion and the Bayesian information criterion, respectively. They are a measure of the model's fit to the data; the lower the value, the better the model.

Table 2 shows the maximum likelihood estimates of the three volatility models for PFE, JNJ, and MRK. Based on the AIC and BIC criteria, we can see that the GARCH-M-L model is the

best model for all three stocks, followed by the GARCH-M model, and then the GARCH model. Moreover, the log-likelihood values also strengthen this conclusion in which the GARCH-M-L models have the highest log-likelihood values. So, it is clear that including the mean and leverage effects improves the model's fit to the data.

Question 5

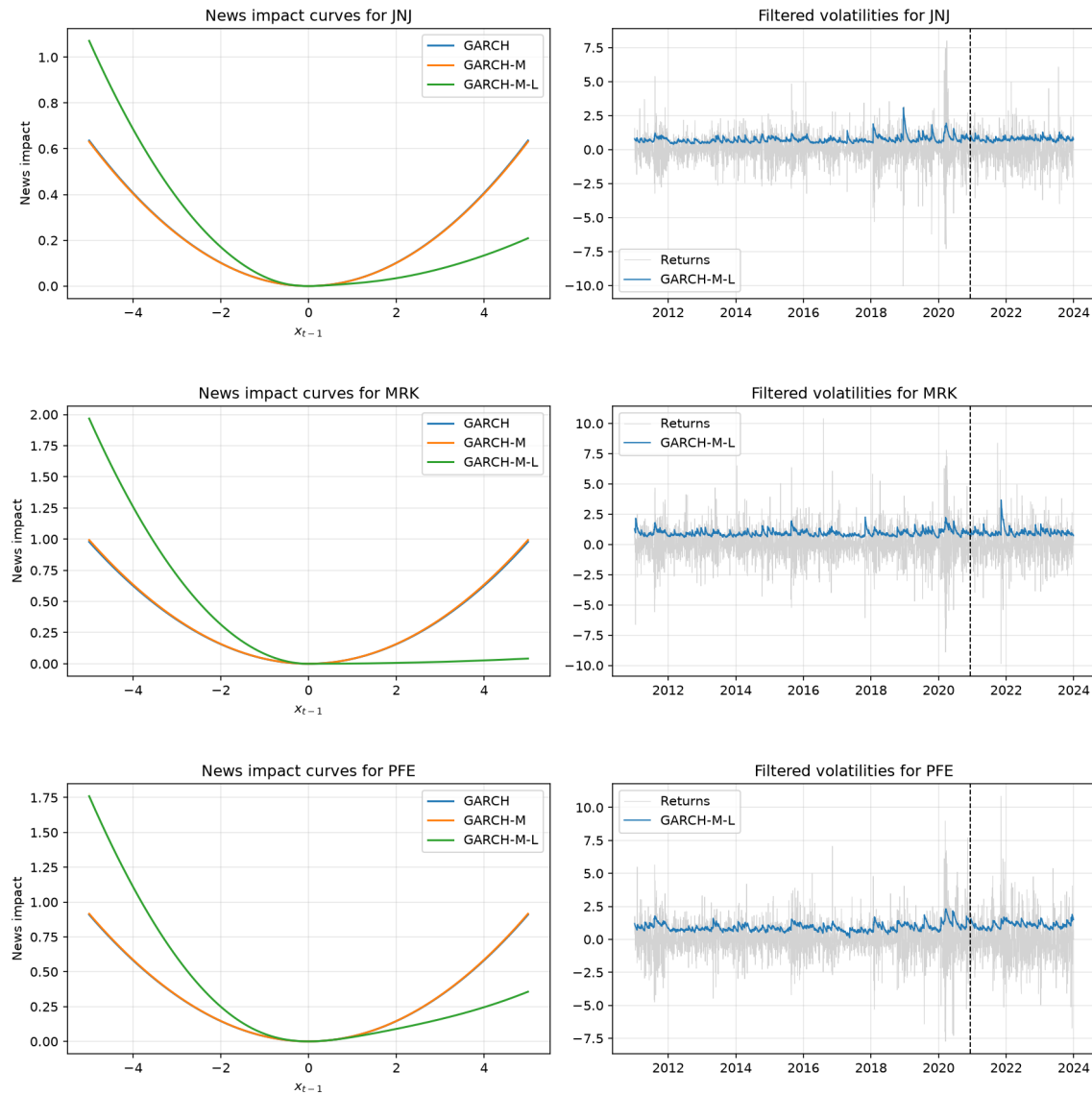


Figure 3: News impact curves of the three models and the filtered volatilities from the GARCH-M-L model for the four stocks.

Question 6

Table 3: Simulation-based Value-at-Risk estimates

Stock	Model	1 step			5 steps			20 steps		
		1%	5%	10%	1%	5%	10%	1%	5%	10%
PFE	M1	-3.78	-2.24	-1.62	-7.93	-4.95	-3.73	-14.47	-9.31	-6.94
PFE	M2	-3.76	-2.19	-1.58	-7.55	-4.74	-3.51	-13.02	-8.12	-5.86
PFE	M3	-3.82	-2.28	-1.63	-7.80	-4.75	-3.45	-13.70	-8.20	-5.80
JNJ	M1	-2.41	-1.42	-1.02	-5.10	-3.13	-2.23	-10.10	-5.57	-3.87
JNJ	M2	-2.50	-1.46	-1.05	-5.29	-3.10	-2.23	-9.29	-5.60	-3.89
JNJ	M3	-2.41	-1.41	-1.00	-5.18	-3.00	-2.16	-9.77	-5.83	-3.91
MRK	M1	-3.00	-1.71	-1.22	-5.92	-3.74	-2.72	-12.02	-7.15	-5.08
MRK	M2	-2.99	-1.71	-1.22	-6.36	-3.82	-2.75	-12.05	-7.39	-5.29
MRK	M3	-3.09	-1.74	-1.27	-6.70	-4.03	-2.90	-12.18	-7.53	-5.45

Notes: The table reports estimated lower-tail Value-at-Risk (VaR) quantiles for the 1-, 5-, and 20-day compound holding-period returns at the 1%, 5%, and 10% probability levels. M1 denotes the GARCH model, M2 the GARCH-M model, and M3 the GARCH-M-L model. Model parameters are estimated using the first 2,500 observations and are kept fixed while the conditional volatility is filtered up to and including January 4, 2021. VaR estimates are based on 10,000 Monte Carlo simulations and are reported in percentage points.